Mahmoud Elsalmy

Computer Engineer

Egypt | **Phone:** +201097128423 | **Email:** mahmoudelsalmy1@gmail.com |
LinkedIn: https://www.linkedin.com/in/mahmoudelsalmy | **GitHub:** https://github.com/mahmoudelsalmy | **Portfolio**

Professional Summary

Computer Systems Engineering student at MSA University with hands-on experience in software development, AI/ML, and embedded systems. Proficient in Python, C/C++, Java, C#, ASP.NET Core MVC, and SQL databases. Experienced in building AI-based systems, web, desktop, and IoT-based systems through academic and practical projects. Seeking an internship or entry-level opportunity to apply problem-solving and software engineering skills in real-world applications.

Education

Bachelor of **Computer Systems Engineering** — 2023 - 2028
October University for Modern Sciences and Arts - **MSA University** — Giza, Egypt

Certification & Training

- **Web Development Using .NET** | *Information Technology Institute (**ITI**) (Link)* — Summer 2025
- **Information Technology Internship** | *EncryptCore Company (Link)* — Aug 2025 – Sep 2025
- **Introduction to Web Technology** | *Information Technology Institute (**ITI**) (Link)* — Summer 2024
- **HCIA-AI** (Artificial Intelligence) | *Huawei (Link)* — Aug 2024 – Sep 2024
- **HCIA-Security** (Security System) | *Huawei (Link)* — Jun 2024 – Jul 2024

Technical Skills

Programming Languages: Python, C, C++, Java, C#
Web Development: ASP.NET Core MVC, Entity Framework, LINQ, HTML5, CSS3, Bootstrap
Databases: SQL Server, MySQL
Embedded Systems & IoT: ESP32, ESP8266, PIC16F877A, Arduino
Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms, Design Patterns
Developer Tools: Git, GitHub, Visual Studio, VS Code, NetBeans, SQL Server Management Studio
Soft Skills: Problem Solving, Teamwork, Continuous learning, Time Management, Communication Skills
Languages: Arabic (Native), English (Intermediate)

Projects

Predictive Maintenance System (Machine Learning) | *Python, Streamlit, Scikit-learn, PCA (Link)*
- Architected a web-based Predictive Maintenance System for equipment fault detection using vibration sensor data with **95%+** classification accuracy.
- Built an end-to-end **machine learning pipeline** including data preprocessing, PCA, and model evaluation.
- Deployed an interactive **Streamlit** web application supporting **single** and **batch** predictions with confidence scores and maintenance recommendations.

Smart Home Automation System (IoT) | *ESP32, Blynk Mobile App, Relay Module, UART (Link)*
- Engineered an IoT-based smart home automation using an ESP32 microcontroller with relay module for **controlling 220V household appliances** (e.g., lighting, fans) via Wi-Fi.
- Integrated the **Blynk mobile application** to provide a user-friendly interface for real-time monitoring and seamless device control.
- Configured relay modules for safe and reliable power switching of high-voltage devices, ensuring system stability and electrical safety.

Automated Plant Watering System (IoT) | *Embedded C , PIC16F877A, ESP8266 (Link)*
- Designed an IoT-based automated irrigation system using PIC16F877A and ESP8266 for **real-time soil moisture** and **water level monitoring**.
- Implemented dual control modes (Automatic/Manual) pump control with safe **relay operation**, **UART communication**, and **LCD display for local status updates**.
- Created a **web dashboard** for remote monitoring featuring data visualization charts to **track plant health**.

Smart Inventory Management System (ERP System) | *C#, ASP.NET Core MVC, SQL Server (Link)*
- Developed a Smart Inventory Management System for efficient stock, product, and user management.
- Implemented **CRUD operations**, role-based access control, and automated low-stock alerts.
- Designed **analytical dashboards** and generated **PDF/Excel** formats with a responsive UI using **Bootstrap**.

University Management System | *Java, JavaFX, MySQL (Link)*
- Developed a Java-based system with **JavaFX GUI** to manage university operations (admin, student, staff) using OOP principles.
- Integrated **MySQL** for persistent data storage and used **JDBC** for database connectivity.